⚗️ ACIDS, BASES AND SALTS

TOPPER NOTES

🧪 Chemistry Chapter 2 | 🔥 Reactivity Masterclass | 🌻 Hinglish Edition

1. Acids and Bases (Introduction) 🍋

- **Acids:** Substances jo taste mein sour (khattay) hote hain aur blue litmus ko red mein convert karte hain. Yeh aqueous solution mein **Hydrogen ions (H^+)** release karte hain.
  Example: HCl, H_2SO_4, CH_3COOH (Vinegar, Lemon).
- **Bases:** Substances jo taste mein bitter (kadwe) aur soapy touch ke hote hain, jo red litmus ko blue mein convert karte hain. Yeh aqueous solution mein **Hydroxide ions (OH^-)** release karte hain (Jo base water mein soluble hote hain, unhe **Alkali** kehte hain).
  Example: $NaOH, KOH, Ca(OH)_2$.

2. Indicators (Suchak) 🎨

Indicators aise substances hote hain jo acid ya base ki presence mein apna colour ya smell change karte hain.

- **Natural Indicators:** Litmus, Turmeric (Haldi), Red Cabbage leaves.
- **Synthetic Indicators:** Phenolphthalein (Acid mein colourless, Base mein pink), Methyl Orange (Acid mein red, Base mein yellow).
- **Olfactory Indicators (Smell-based):** Onion, Vanilla, Clove oil (Acid mein smell rehti hai, base mein smell destroy ho jati hai).

3. Chemical Properties of Acids and Bases ⚡

A. Action with Metals:

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Acid + Metal $\rightarrow$ Salt + Hydrogen gas ($H_2$) Example: $Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$ *(Test for $H_2$ gas: Burns with a pop sound!)*
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B. Action with Metal Carbonates / Hydrogen Carbonates:

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Acid + Metal Carbonate $\rightarrow$ Salt + Water + Carbon Dioxide ($CO_2$) Example: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2 \uparrow$ *(Test for $CO_2$ gas: Turns lime water milky when passed through it due to insoluble $CaCO_3$)*
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C. Neutralisation Reaction (Amla aur Khaar ki Abhikriya):

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Acid + Base $\rightarrow$ Salt + Water Example: $HCl + NaOH \rightarrow NaCl + H_2O$
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💡 **Metal Oxides vs Non-Metal Oxides:** Metal oxides basic in nature hote hain (jaise CaO), jabki non-metal oxides acidic in nature hote hain (jaise CO_2).

4. The pH Scale (Power of Hydrogen) 📊

Definition: A scale for measuring hydrogen ion concentration in a solution is called pH scale (range from 0 to 14).

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Text-Based pH Scale Diagram: [ 0 <------------------ 7 ------------------> 14 ] Strong Acid | Neutral ($H_2O$) | Strong Base (Red / Orange) | (Green / Pure) | (Blue / Purple) | pH = 7 (Neutral)
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Importance of pH in Daily Life:

- **Our Body:** Human body pH range 7.0 to 7.8 ke beech work karti hai.
- **Tooth Decay:** Mouth ka pH 5.5 se kam hone par enamel (tooth coating) decay hone lagta hai.
- **Stomach Acid:** Stomach mein excess HCl banne par acidity hoti hai, jise neutralize karne ke liye **Antacids** (jaise Milk of Magnesia) use kiye jate hain.
- **Bee Sting / Ant Sting:** Cheenti ya madhumakhi ke dank mein formic acid hota hai, jisse dard hota hai. Relief ke liye mild base (jaise baking soda) apply kiya jata hai.

5. Important Chemical Salts 🧂

A. Common Salt (Sodium Chloride - $NaCl$)

Derived from sea water; raw material for many chemicals.

B. Sodium Hydroxide (Caustic Soda - $NaOH$) [Chlor-Alkali Process]

Electricity pass ki jati hai aqueous solution of $NaCl$ (brine) mein:

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$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$ *(Byproducts: Chlorine gas at anode, Hydrogen at cathode, NaOH near cathode)*
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C. Bleaching Powder ($CaOCl_2$)

Prepared by action of chlorine on dry slaked lime: $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$.

D. Baking Soda ($NaHCO_3$)

Mild non-corrosive basic salt: $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$.

E. Washing Soda ($Na_2CO_3 \cdot 10H_2O$) & Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$)

6. 🌟 Dedicated Chemical Formulas & Compounds Corner

1. Common Salt: NaCl 2. Sodium Hydroxide: NaOH (Caustic Soda) 3. Bleaching Powder: CaOCl_2 (Calcium oxychloride) 4. Baking Soda: NaHCO_3 (Sodium hydrogen carbonate) 5. Washing Soda: Na_2CO_3 \cdot 10H_2O (Sodium carbonate decahydrate) 6. Plaster of Paris: CaSO_4 \cdot 1/2H_2O (Calcium sulphate hemihydrate) 7. Gypsum: CaSO_4 \cdot 2H_2O (Calcium sulphate dihydrate) 8. Washing Soda conversion: Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O

7. Differences: Acids vs. Bases ⚖️

Feature	Acids 🍋	Bases 🧼
Taste	Sour (Khattay)	Bitter (Kadwe)
Litmus Test	Turns Blue litmus to Red	Turns Red litmus to Blue
Ions produced	H^+ ions in water	OH^- ions in water

8. Exam Short Tricks & Tips 🚀

- **POP Sound Trick:** H haraf se Hydrogen, H se Hi Pop sound! (H_2 gas test).
- **Water of Crystallization trick:** Plaster of Paris mein **half ($1/2$) water molecule** hota hai, jabki Gypsum mein **two (2) water molecules** hote hain! ($CaSO_4 \cdot 1/2H_2O$ vs $CaSO_4 \cdot 2H_2O$).

9. ⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Acid:** Releases H^+, pH < 7.
- [ ] **Base:** Releases OH^-, pH > 7.
- [ ] **Chlor-Alkali process:** Produces $NaOH, Cl_2, H_2$.
- [ ] **Plaster of Paris formula:** $CaSO_4 \cdot 1/2H_2O$.

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